

Supplement to Repair What Matters

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1 What this supplement adds

The main paper contains the revision contract, the full paper-grade campaign, and the residual-sensitive confirmation. This supplement collects three details that are useful for review but awkward to fit into the main narrative: how calibrated work should be interpreted, representative trained-3DGS decision cases, and the display-space bound used by the residual-sensitive mechanism study.

2 Calibrated work and practical benefit

The planner compares a candidate local repair with a frozen full-rebuild baseline using

$$W(C) = \sum_d \kappa_d n_d(C).$$

Here $n_d(C)$ is a native counter and κ_d is the pre-campaign conversion factor for that counter’s domain. The paper-grade campaign calibrates Gaussian inspection, Gaussian update, GPU-publication bytes, and temporal-invalidated pixels. This produces a common work scale without treating those native operations as interchangeable physical events.

A ratio below one means the selected repair accounts for less work than the independently measured full baseline on this frozen scale. A ratio of one denotes FULL fallback. The metric deliberately excludes wall-clock effects such as allocator behavior, cache locality, synchronization, command submission, and GPU overlap.

Table 1: Paper-grade calibrated-work interpretation by dataset/representation. “Work avoided” is $1 - \text{median}(W/W_{\text{full}})$.

Dataset / representation	Median work/FULL	Median work avoided
3RScan	0.341	65.94%
ARKitScenes	0.483	51.68%
Bonn RGB-D Dynamic	0.475	52.45%
Trained GraphDECO 3DGS	0.519	48.09%
Overall	0.474	52.55%

3 Representative trained-3DGS behavior

The next figures use a pinned pretrained GraphDECO 3DGS scene from the frozen evaluation. They are revision-fidelity visualizations: the selected repair is compared with a full-after result inside the same representation.

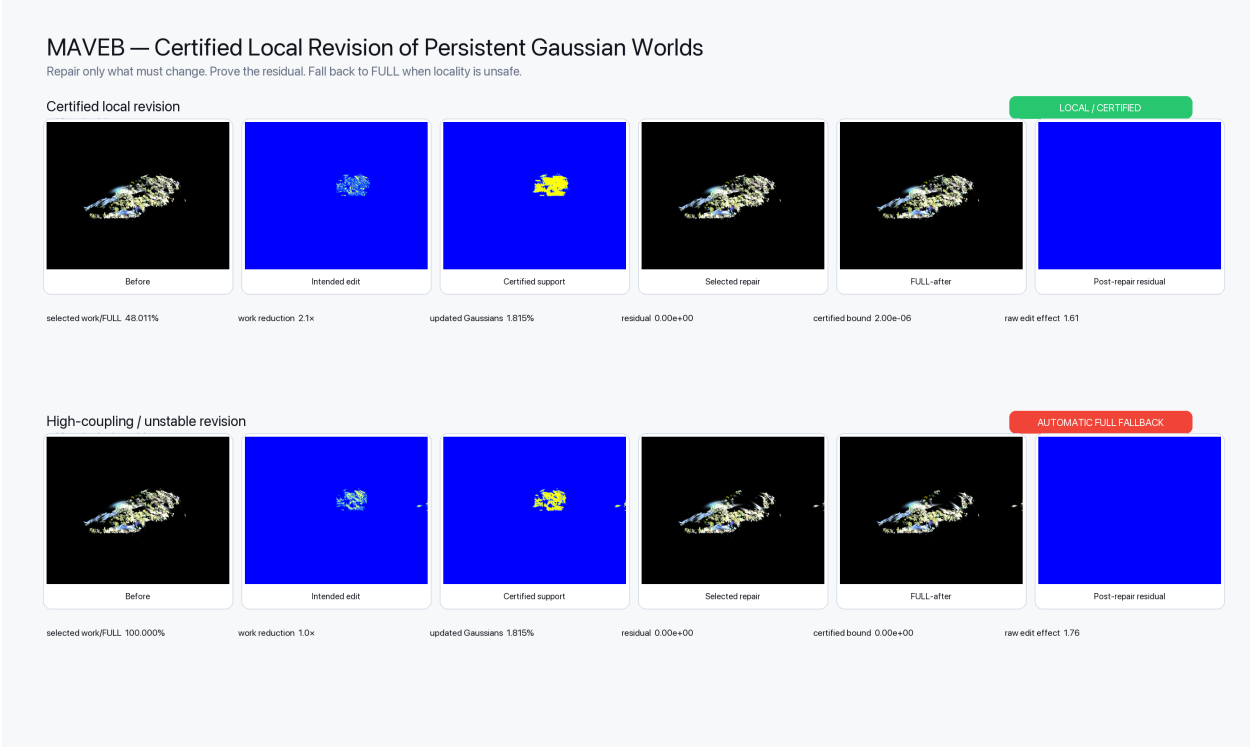


Figure 1: A certified local case and an automatic full fallback on the same trained-3DGS representation. The local case records work/FULL=48.011%; the fallback records 100%.

4 Residual-sensitive evidence chronology

The broad 1,275-case campaign reproduces the independently generated full-after raster exactly at audited 8-bit precision, so it cannot test whether a non-zero local residual can still be safe. The residual-sensitive experiments are frozen separately.

Supplement: Gaussian finite-edit display bound

For one protected pixel, let U be the unchanged ordered Gaussian support and E the complete edited support. Effective per-splat alpha is assumed to be post-clamp/post-cutoff and to satisfy $\alpha_i \in [0, 1]$. Per-channel emitted color after spherical-harmonic evaluation and the background lie in the same interval of width C_{color} .

For edited support E , define

$$A(E) = 1 - \prod_{i \in E} (1 - \alpha_i).$$

This is the maximum fraction of final transmittance whose contribution can be replaced, attenuated, or revealed through the edited splats. Since any channel value differs by at most C_{color} ,

$$\|R(U \cup E) - R(U)\|_{\infty} \leq C_{\text{color}} A(E).$$

For before/after edited sets E_0, E_1 , insert the common unchanged state U and apply the triangle

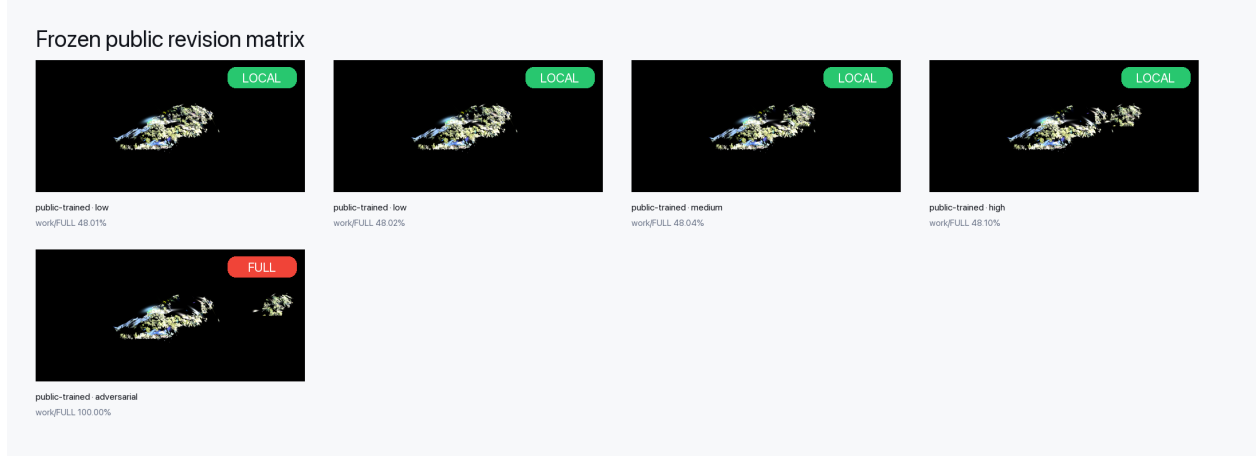


Figure 2: Frozen coupling sweep on a trained-3DGS scene. Low, medium, and one high-coupling case remain local at roughly 48% of the calibrated full baseline, while the adversarial case selects FULL.

Table 2: Residual-sensitive sequence retained in the final claim.

Stage	Population	Result
v4 diagnostic	4 scenes, 512 cases	Every non-zero-residual case falls back to FULL; this exposes the magnitude-insensitive display certificate as the bottleneck.
v5 mechanism	4 scenes, 768 cases	The opacity-only delta-sensitive certificate yields 142 certified non-zero local cases and 29 tolerance crossovers; the unchanged translation control yields zero.
v6 confirmation	20 scenes, 3,840 pooled parameter cases	Non-zero opacity locality and tolerance crossover occur in all 20 scenes. The scene-level opacity-minus-translation difference is 0.477 with dataset-stratified bootstrap 95% interval [0.432, 0.517].

inequality:

$$\begin{aligned}
\|R(U \cup E_0) - R(U \cup E_1)\|_\infty &\leq \|R(U \cup E_0) - R(U)\|_\infty \\
&\quad + \|R(U) - R(U \cup E_1)\|_\infty \\
&\leq C_{\text{color}} (A(E_0) + A(E_1)).
\end{aligned}$$

The output channel itself cannot change by more than its admissible interval width, hence

$$\|R(U \cup E_0) - R(U \cup E_1)\|_\infty \leq C_{\text{color}} \min\{1, A(E_0) + A(E_1)\}.$$

Insertion and deletion are obtained by taking one edited set to be empty. Depth-order changes involving edited splats are included because the edited set must contain every Gaussian whose projected opacity, color, or order relation can change. Renderer behavior outside these bounded assumptions is not softened into the certificate; it becomes an exact dependency or triggers FULL fallback. The image-space certificate is the maximum of the pixel bound over the declared protected support.

MAVEB evidence dashboard

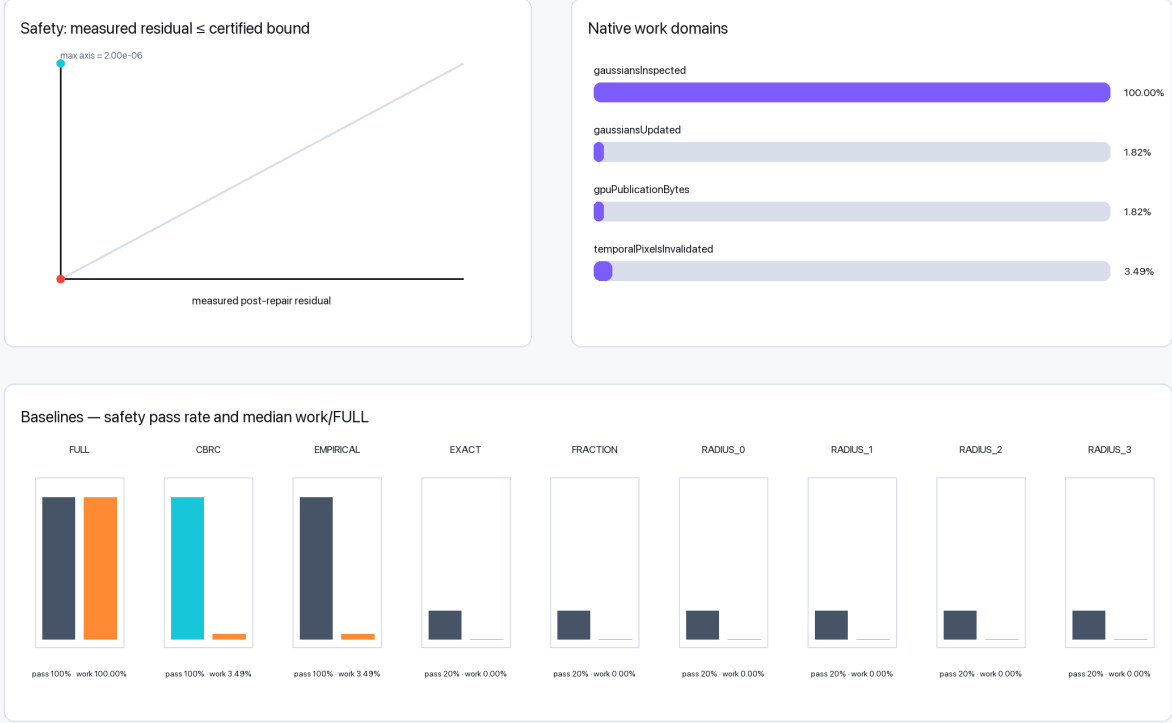


Figure 3: Representative evidence dashboard exposing the safety check, native work domains, and baseline behavior for a trained-3DGS case.

Opacity-only delta refinement used by v5–v6

The general finite-edit bound above is deliberately indifferent to edit magnitude once the edited support is known. This is safe for geometry, color, insertion, deletion, and ordering changes, but it can be unnecessarily loose when corresponding Gaussian records differ only in opacity. The confirmatory v5–v6 path therefore uses a narrower bound under stronger invariants.

Consider front-to-back compositing at one protected pixel. Let $\alpha_i^0, \alpha_i^1 \in [0, 1]$ be the effective renderer-admitted alpha values for paired Gaussian i before and after the revision. Position, projected covariance, source-order depth, rotation, and spherical-harmonic color are identical for each pair. Denote the accumulated color after processing splats $1, \dots, n$ by C_n . Changing one alpha coordinate while holding all other coordinates fixed can replace at most the remaining channel range multiplied by the alpha change. If every possible emitted/background channel lies in an interval of width C_{color} , then

$$\|C_n(\dots, \alpha_i^0, \dots) - C_n(\dots, \alpha_i^1, \dots)\|_{\infty} \leq C_{\text{color}} |\alpha_i^0 - \alpha_i^1|.$$

Applying this coordinate-wise along a path from the full before alpha vector to the full after alpha vector and using the triangle inequality gives

$$\Delta_p \leq C_{\text{color}} \sum_i |\alpha_i^0(p) - \alpha_i^1(p)|.$$

The renderer used by the independent oracle may stop compositing once accumulated alpha exceeds 0.999. Relative to the corresponding non-terminated compositor, either endpoint can then omit at most 0.001 transmittance. A before/after difference therefore needs at most an additional $0.002C_{\text{color}}$ allowance. The implemented pixel certificate is

$$B_p^{\text{opacity}} = C_{\text{color}} \min \left(1, \sum_i |\alpha_i^0(p) - \alpha_i^1(p)| + 0.002 \right)$$

whenever at least one admitted alpha changes at p . If the summed alpha variation is exactly zero, the implementation returns an exact zero bound instead of adding the early-termination allowance globally. This preserves zero support outside the affected pixels.

The refinement is intentionally fail-closed. Paired primitive counts and spherical-harmonic degree must agree, and every paired Gaussian must be identical except for opacity logit. A geometry, covariance, rotation, spherical-harmonic, or unexpected visibility change rejects the refinement and returns control to the general finite-edit certificate or to full fallback. The v5 mechanism experiment and v6 confirmatory breadth experiment use this opacity-only path without changing the general translation control.

5 Claim boundaries

The calibrated-work result is not a wall-clock speedup claim. Selected-to-FULL raster equality measures revision fidelity in the evaluated representation and camera, not reconstruction quality against source photographs. The planner does not claim globally minimum-work cones. The 3,840 v6 parameter cases are pooled evidence; scene-level, dataset-stratified inference is the statistical unit. The opacity-only refinement is valid only under the paired-record invariants stated in the derivation, and unsupported cases fail closed.

6 Ethics Statement

N/A. This work reports software, rendering, and reconstruction experiments and does not recruit human participants or collect personal data from participants. Evaluation uses public research datasets and pretrained research assets under their upstream terms; the repository does not redistribute the upstream datasets.

7 Generative AI Use Disclosure

OpenAI ChatGPT was used as an assistive tool for literature-search support, software and scripting assistance, debugging, experimental planning, visualization and manuscript tooling, and drafting or language editing. Reported measurements and frozen evidence were produced by the project software and experimental artifacts. The author reviewed the final technical text and remains responsible for the code, citations, analysis, claims, conclusions, and final manuscript.